Field of a Dipole in Coordinate-free Form

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Coordinate-free Form

All the equations we will discuss in this note:

$$\begin{cases} \mathbf{E}_{\text{dip}}(\mathbf{r}) = \dfrac{1}{4\pi\epsilon_0}\dfrac{1}{r^3}\left[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}\right] \\ \mathbf{B}_{\text{dip}}(\mathbf{r}) = \dfrac{\mu_0}{4\pi}\dfrac{1}{r^3}\left[3(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}\right] \end{cases}$$

1 Electric Field of a (Perfect) Dipole

The electric dipole potential $V_{\text{dip}}(\mathbf{r})$ is

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}, \tag{1}$$

where $\mathbf{p}$ is the dipole moment of the distribution which depends on the *source coordinates*:

$$\mathbf{p} \coloneqq \int \mathbf{r}'\varrho(\mathbf{r}')\,d\tau'. \tag{2}$$

The electric field of a perfect dipole satisfies that

$$\mathbf{E}_{\text{dip}}(\mathbf{r}) = -\boldsymbol{\nabla} V_{\text{dip}}(\mathbf{r}) \tag{3}$$

We can evaluate the gradient of the potential, that is

$$\begin{aligned} \boldsymbol{\nabla} V_{\text{dip}}(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0}\boldsymbol{\nabla}\left[\frac{1}{r^2}(\mathbf{p}\cdot\hat{\mathbf{r}})\right] = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^2}\Big(\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}})\Big) + (\mathbf{p}\cdot\hat{\mathbf{r}})\left(\boldsymbol{\nabla}\frac{1}{r^2}\right)\right] \\ &= \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^2}\Big(\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}})\Big) - \frac{2(\mathbf{p}\cdot\hat{\mathbf{r}})}{r^3}\hat{\mathbf{r}}\right] \end{aligned} \tag{4}$$

And the term $\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}})$ can be expressed to

$$\begin{aligned} [\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}})]_i &= \Big[\mathbf{p}\times(\boldsymbol{\nabla}\times\hat{\mathbf{r}}) + \hat{\mathbf{r}}\times(\boldsymbol{\nabla}\times\mathbf{p}) + (\mathbf{p}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}} + (\hat{\mathbf{r}}\cdot\overset{\frown}{\boldsymbol{\nabla})\mathbf{p}}\Big]_i \\ &= \Big[(\mathbf{p}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}}\Big]_i = p_j\partial_j\left(\frac{x_i}{r}\right) = \frac{p_j\delta_{ij}}{r} - \frac{p_jx_ix_j}{r^3} = \frac{p_i}{r} - \frac{(\mathbf{p}\cdot\mathbf{r})}{r^3}x_i. \end{aligned}$$

Then $\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}})$ results in

$$\boldsymbol{\nabla}(\mathbf{p}\cdot\hat{\mathbf{r}}) = \frac{\mathbf{p}}{r} - \frac{(\mathbf{p}\cdot\mathbf{r})\mathbf{r}}{r^3} = \frac{\mathbf{p}}{r} - \frac{(\mathbf{p}\cdot\hat{\mathbf{r}})}{r}\hat{\mathbf{r}} = \frac{1}{r}[\mathbf{p} - (\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}]$$

Substituting the result into (4), it becomes

$$\boldsymbol{\nabla}V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^3}\Big(\mathbf{p} - (\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}\Big) - \frac{2(\mathbf{p}\cdot\hat{\mathbf{r}})}{r^3}\hat{\mathbf{r}}\right] = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\big[\mathbf{p} - 3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}\big], \tag{5}$$

Remember to take the minus sign; we can finally get the electric field

$$\boxed{\mathbf{E}_{\text{dip}}(\mathbf{r}) = -\boldsymbol{\nabla}V = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}\right]} \tag{6}$$

2 Magnetic Field of a (Perfect) Dipole

The magnetic dipole vector potential $\mathbf{A}_{\text{dip}}(\mathbf{r})$ is defined as:

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{\mathbf{r}}}{r^2}, \tag{7}$$

where $\mathbf{m}$ is the magnetic dipole moment, that is

$$\mathbf{m} \coloneqq I\int d\mathbf{a}' \tag{8}$$

The magnetic field of a perfect dipole satisfies that

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \boldsymbol{\nabla}\times\mathbf{A}_{\text{dip}}(\mathbf{r}) \tag{9}$$

Substituting into (7), we have

$$\boldsymbol{\nabla}\times\mathbf{A} = \frac{\mu_0}{4\pi}\boldsymbol{\nabla}\times\left(\frac{\mathbf{m}\times\hat{\mathbf{r}}}{r^2}\right) = \frac{\mu_0}{4\pi}\left[\frac{1}{r^2}\boldsymbol{\nabla}\times(\mathbf{m}\times\hat{\mathbf{r}}) - (\mathbf{m}\times\hat{\mathbf{r}})\times\boldsymbol{\nabla}\left(\frac{1}{r^2}\right)\right]$$

$$= \frac{\mu_0}{4\pi}\frac{1}{r^2}\left[(\hat{\mathbf{r}}\cdot\boldsymbol{\nabla})\mathbf{m} - (\mathbf{m}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}} + \mathbf{m}(\boldsymbol{\nabla}\cdot\hat{\mathbf{r}}) - \hat{\mathbf{r}}(\boldsymbol{\nabla}\cdot\mathbf{m})\right] + \frac{\mu_0}{4\pi}\frac{2(\mathbf{m}\times\hat{\mathbf{r}})\times\hat{\mathbf{r}}}{r^3}.$$

We can first expand the terms that

$$\begin{cases} (\mathbf{m}\times\hat{\mathbf{r}})\times\hat{\mathbf{r}} = -\hat{\mathbf{r}}\times(\mathbf{m}\times\hat{\mathbf{r}}) = -\mathbf{m}(\hat{\mathbf{r}}\cdot\hat{\mathbf{r}}) + \hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{m}) = (\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m} \\ \big[-(\mathbf{m}\cdot\boldsymbol{\nabla})\hat{\mathbf{r}}\big]_i = -m_j\partial_j\dfrac{x_i}{r} = -\dfrac{m_j\delta_{ij}}{r} + \dfrac{m_jx_jx_i}{r^3} = -\dfrac{m_i}{r} + \dfrac{(\mathbf{m}\cdot\mathbf{r})x_i}{r^3} \\ \big[\mathbf{m}(\boldsymbol{\nabla}\cdot\hat{\mathbf{r}})\big]_i = \partial_j\dfrac{x_j}{r}m_i = \dfrac{3m_i}{r} - \dfrac{x_jx_jm_i}{r^3} = \dfrac{2m_i}{r}. \end{cases} \tag{10}$$

By (10), the curl of the vector potential (2) is

$$\boldsymbol{\nabla} \times \mathbf{A} = \frac{\mu_0}{4\pi}\left[\frac{1}{r^2}\left(\frac{\mathbf{m}}{r} + \frac{(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}}{r}\right) + \frac{2(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - 2\mathbf{m}}{r^3}\right],$$

this can be simplified as the final form of the magnetic field that

$$\boxed{\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[3(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}\right]} \tag{11}$$

The expressions (6) & (11) decompose the dipole field into a radial component and a transverse component relative to the observation direction $\hat{\mathbf{r}}$, making the geometric structure of the dipole field explicit.

Some Identities

Triple Products:

$$\begin{cases} \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}) \\ \mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

Products Rules:

$$\begin{cases} \boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla}g) + g(\boldsymbol{\nabla}f) \\ \boldsymbol{\nabla}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\boldsymbol{\nabla}\times\mathbf{B}) + \mathbf{B}\times(\boldsymbol{\nabla}\times\mathbf{A}) + (\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} + (\mathbf{B}\cdot\boldsymbol{\nabla})\mathbf{A} \\ \boldsymbol{\nabla}\cdot(f\mathbf{A}) = f(\boldsymbol{\nabla}\cdot\mathbf{A}) + \mathbf{A}\cdot\boldsymbol{\nabla}f \\ \boldsymbol{\nabla}\times(f\mathbf{A}) = f(\boldsymbol{\nabla}\times\mathbf{A}) - \mathbf{A}\times\boldsymbol{\nabla}f \\ \boldsymbol{\nabla}\times(\mathbf{A}\times\mathbf{B}) = (\mathbf{B}\cdot\boldsymbol{\nabla})\mathbf{A} - (\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} + \mathbf{A}(\boldsymbol{\nabla}\cdot\mathbf{B}) - \mathbf{B}(\boldsymbol{\nabla}\cdot\mathbf{A}) \end{cases}$$

Second Derivatives:

$$\begin{cases} \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0 \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0} \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{cases}$$

Green's Identities and Some Integrals:

$$\begin{cases} \displaystyle\int_{\mathcal{V}} \left(T\,\nabla^2 U - U \nabla^2 T\right) d\tau = \oint_{\mathcal{S}} \left(T \boldsymbol{\nabla} U - U \boldsymbol{\nabla} T\right) \cdot d\mathbf{a} \\ \displaystyle\int_{\mathcal{S}} \boldsymbol{\nabla} T \times d\mathbf{a} = -\oint_{\mathcal{P}} T d\boldsymbol{\ell} \\ \displaystyle\oint (\mathbf{c} \cdot \mathbf{r})\, d\boldsymbol{\ell} = \left(\int d\mathbf{a}\right) \times \mathbf{c} \\ \displaystyle\int_{\mathcal{V}} \boldsymbol{\nabla} \times \mathbf{F}\, d\tau = -\oint_{\mathcal{S}} \mathbf{F} \times d\mathbf{a} \end{cases}$$

Delta functions:

$$\begin{cases} \boldsymbol{\nabla} \cdot \left(\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi \delta^3(\boldsymbol{\mathscr{r}}) \\ \boldsymbol{\nabla} \left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla} \left(\dfrac{1}{\mathbf{r} - \mathbf{r}'}\right) = -\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \boldsymbol{\nabla}' \left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}' \left(\dfrac{1}{\mathbf{r} - \mathbf{r}'}\right) = \dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \nabla^2 \left(\dfrac{1}{\mathscr{r}}\right) = -4\pi \delta^3(\boldsymbol{\mathscr{r}}) \end{cases}$$